INCH-POUND
MIL-V-17547A(SH)
9 November 1988
SUPERSEDING
MIL-V-17547(NAVY)
28 August 1953
(See 6.7)

MILITARY SPECIFICATION

VALVES, CHECK, BRONZE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes check valves of various types for low pressure air, water, and oil services.

1.2 Classification. Check valves shall be of the following types and classes as specified (see 6.1 and 6.2.1):

Type A - Swing
Type B - Lift horizontal
Type C - Lift vertical
Type D - Lift angle
Type E - Tilting disc (top entry)
Type F - Tilting disc (in line)
Type G - Split disc (wafer type)
Type H - Split disc (lug type)
Class 1 - For working pressure 50 pounds per square inch (lb/in^2) max., 14 inches and larger (for types E, F, G and H).
Class 2 - For working pressure 250 lb/in^2 max., 2-1/2 to 12 inches (for types A, B, C, D, G and H).
Class 3 - For working pressure 400 lb/in^2 max., 1/4 to 2 inches (for types A, B, C and D).
Class 4 - For working pressure 400 lb/in^2 max., 2-1/2 to 10 inches (for types A, B, C and D).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

1.2.1 The following requirements pertain to type A class 2 and 3 valves shown on Drawings 803-1385637 and 803-1385721 respectively. The drawings specify all of the requirements for the valves depicted and do not depend on any of the requirements contained in this specification. Conversely the requirements in this specification do not depend on any of the requirements in the drawings.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-N-281	- Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286	- Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
QQ-N-288	- Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-C-591	- Copper-Silicon, Copper-Zinc-Silicon, and Copper-Nickel-Silicon Alloys: Rod, Wire, Shapes, Forgings, and Flat Products (Flat Wire, Strip, Sheet, Bar and Plate).

MILITARY

MIL-V-3	- Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-G-1149	- Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness.
MIL-F-1183	- Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for.
MIL-C-6183	- Cork and Rubber Composition Sheet; for Aromatic Fuel and Oil Resistant Gaskets.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-F-20042	- Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-L-24131	- Lubricant, Colloidal Graphite in Isopropanol.
MIL-B-24480	- Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

MILITARY (Continued)

MIL-S-45180 - Sealing Compound, Gasket, Hydrocarbon Fluid and Water Resistant.
MIL-R-83248 - Rubber, Fluorocarbon Elastomer, High-Temperature, Fluid, and Compression Set Resistant.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-798 - Non-Destructive Testing, Welding, Quality Control, Material Control and Identification and Hi-Shock Test Requirements for Piping System Components for Naval Shipboard Use.

2.1.2 Government drawings and publication. The following Government drawings and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 803-1385637 - Valves, Bronze, Flanged, Check 250 PSI WOG.
NAVSHIPS 803-1385721 - Valves, Bronze, 1/4-2 Union End Swing Check, 400 PSI WOG.

PUBLICATION

NAVSEA

0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.24 - Bronze Pipe Flanges and Flanged Fittings. (DoD adopted)
B18.2.1 - Square and Hex Bolts and Screws Inch Series Including Hex Cap Screws and Lag Screws. (DoD adopted)
B18.2.2 - Square and Hex Nuts (Inch Series). (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 61 - Standard Specification for Steam or Valve Bronze Castings. (DoD adopted)
B 98 - Standard Specification for Copper-Silicon Alloy Rod, Bar, and Shapes. (DoD adopted)
B 148 - Standard Specification for Aluminum-Bronze Sand Castings. (DoD adopted)
B 164 - Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire. (DoD adopted)
B 443 - Standard Specification for Nickel-Chromium-Molybdenum-Columbium Alloy (UNS N06625) Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

HS J1086 - Metals and Alloys in the Unified Numbering System. (DoD adopted)
AMS 4676 - Alloy Bars and Forgings, Corrosion Resistant.
AMS 4892 - Alloy Castings, Sand and Centrifugal, Corrosion and Heat Resistant.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials. Unless otherwise specified (see 6.2.1), valve materials shall be in accordance with table I. Materials other than those specified in table I may be used only upon approval of NAVSEA. A certificate of compliance shall be prepared (see 6.2.2 and appendix C).

TABLE I. List of material.

Valve parts	Material	Specification or standard	Remarks
Body Cap or cover Hanger or lever Disc	Valve bronze	QQ-C-390, ASTM B 61 (UNS C92200) for sizes under 16 inches only	See notes 1/ and 2/
	Nickel-aluminum-bronze	MIL-B-24480, ASTM B 148 alloy 958 (UNS C95800), all sizes	
Seat Hinge pin Disc nut Disc nut pin Disc washer Lever plug	Nickel-copper alloy	QQ-N-281, ASTM B 164 (UNS N04400) QQ-N-286; AMS 4676 (UNS N05500), QQ-N-288 Comp. D, AMS 4892 (UNS N04019)	See notes 1/, 2/ and 3/
	Valve bronze	QQ-C-390, ASTM B 61 (UNS C92200)	
Threaded fasteners	Silicon bronze	QQ-C-591, ASTM B 98 (UNS C65100)	ANSI B18.2.1 ANSI B18.2.2 See note 1/
	Nickel-copper alloy	QQ-N-281, ASTM B 164 (UNS N04400)	
Gasket	Nitrile and cork sheet	MIL-C-6183, class 1, grade C firm	For JP-5
	Fluorocarbon	MIL-R-83248, type II, class 1	For oil and air service
	Synthetic rubber	MIL-G-1149, type I, class 2	For water service

1/ Table I contains the commercial and Military specifications and standards, and in addition, contains the corresponding SAE HS J1086 Unified Numbering System (UNS) designation. Reference to a specific UNS designation will permit the use of any one of the many alloy specifications and standards listed in SAE HS J1086 for that UNS number.

2/ Valves for seawater service shall be provided with nickel-copper alloy trim, such as discs, seats, disc nuts, hinge pins, side plugs and washers as applicable.

3/ For valves where low permeability (non-magnetic) material is required, QQ-N-286 shall be used for valve trim.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Service rating. Maximum gauge pressure and maximum temperature for valves shall be as specified in table II.

TABLE II. Service rating.

Type	Class	Nominal pipe size (nps) (inches)	Maximum gauge pressure (lb/in^2)	Maximum temperature (°F)
A	2	2-1/2 to 12 inclusive; 250 lb/in^2 maximum	See Dwg 803-1385637	
	3	1/4 to 2 inclusive; 400 lb/in^2 maximum	See Dwg 803-1385721	
	4	2-1/2 to 10 inclusive; 400 lb/in^2 maximum	400 360 320 280 240 200 150	1/150 200 250 300 350 400 425
B, C and D	2	2-1/2 to 12 inclusive; 250 lb/in^2 maximum	250 235 215 200 180 160 150	1/150 200 250 300 350 400 425
	3	1/4 to 2 inclusive; 400 lb/in^2 maximum	400 360 320 280 240 200 150	1/150 200 250 300 350 400 425
	4	2-1/2 to 10 inclusive; 400 lb/in^2 maximum		
E, F, G and H	1	14 and larger 50 lb/in^2 maximum	50	1/150

See footnote at top of next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

1/ The service pressure rating at temperatures less than 150 degrees Fahrenheit (°F) shall be the same as the rating shown for 150°F. Valves used at these lower temperatures shall conform to the requirements applicable to the condition under which it is used.

3.4 Construction.

3.4.1 Body walls. The walls of valve bodies shall not have flat surfaces. Ribbing shall be provided as necessary to prevent distortion.

3.4.1.1 Threads on threaded parts shall comply with FED-STD-H28.

3.4.2 Discs. Discs shall be either of cast or forged nickel-copper alloy or bronze with an insert of nickel-copper alloy (cast, forged or sheet), in accordance with table I, securely silver brazed to the disc to form the seating surface. Silver brazing shall be in accordance with NAVSEA 0900-LP-001-7000. The finished thickness of the insert shall be at least 1/8 inch for valves 2-1/2 to 6 inches, 5/32 inch for valves 7 to 12 inches, 3/16 inch for valves 14 to 22 inches, 1/4 inch for valves 24 to 30 inches and 3/8 inch for valves 32 to 40 inches.

3.4.3 Seat. Segmented seat rings shall not be permitted.

3.4.4 Body. Valve bodies shall be one-piece construction.

3.4.5 End connections. Unless otherwise specified (see 6.2.1), valves in sizes 2 inches and below shall be furnished with silver-brazing union ends in accordance with MIL-F-1183. Valves in sizes 2-1/2 inches and larger shall be furnished with flanged ends. The flanges shall be cast integrally with the valve body, and the inlet and outlet flange shall be of the same size and pressure rating.

3.4.6 Flanges. Flanges shall be in accordance with MIL-F-20042 using corresponding pressure ratings. Unless otherwise specified (see 6.2.1), valves shall be supplied with flanges drilled in accordance with MIL-F-20042; except where an odd number of bolt holes is specified (see 6.2.1). The flange bolt holes shall be equally spaced and positioned such that the full section cutting plane bisects the angle between the holes. Flanges shall be spot faced or finished on the back parallel to the flange face within 1 degree for the bolt heads and nuts. The spot face shall not cut more than 1/32 inch below the minimum plate thickness. Eccentricity of the spot face diameter with the bolt hole shall be minimized to provide facing for the bolt head. Diameter for spot facing shall be 110 to 120 percent of the bolt head corner-to-corner dimension. Rims need not be finished provided they do not have surface discontinuities such as cracks, sand inclusions, gas holes, cracks or excessive shrinkage.

3.4.7 Face to face dimension. Flanged valves shall have face to face and face to end dimensions as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

3.4.8 Type A, class 4 valves (swing check). Type A, class 4 (2-1/2 to 12 inches, 400 lb/in^2 maximum) valves shall permit an unobstructed flow and the area at any point within the valve in the line of flow shall be equal to the inside diameter of the standard nps. The valves shall operate satisfactorily when installed in either the vertical or horizontal position.

3.4.9 Types B, C and D valves (lift check). Lift check valves shall be of a design with body configuration as specified (see 1.2).

3.4.10 Types E and F valves (tilting disc check). Types E and F valves shall be as specified in 3.4.10.1 through 3.4.10.3.

3.4.10.1 The valves shall incorporate a dash pot device which will provide a cushioned closure for minimum slam and maximum tightness.

3.4.10.2 Valves shall be installed in the horizontal run of the pipe so that the hinge pin is above the horizontal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

TABLE III. Dimensions and weights of flanged valves.

Valve size (nps) (inches)	Types A, B, C Face to face (inches)	Type D Face to end (inches)	Types A, B, C, D Maximum weight (pounds)	Type E		Type F		Types G & H Face to face (inches)	Type G Maximum weight (pounds)	Type H Maximum weight (pounds)
				Face to face (inches)	Maximum weight (pounds)	Face to face (inches)	Maximum weight (pounds)			
2-1/2	10	6-1/4	48	---	---	---	---	2-5/8	11	19
3	10-3/4	6-3/4	58	---	---	---	---	2-7/8	15	38
1/ 3-1/2	11-1/2	---	68	---	---	---	---	---	---	---
4	12-1/4	7-3/4	78	---	---	---	---	2-7/8	18	48
1/ 4-1/2	13	---	88	---	---	---	---	---	---	64
5	13-3/4	8-3/4	110	---	---	---	---	3-3/8	35	80
6	15-1/4	9-1/2	150	---	---	---	---	3-7/8	45	96
1/ 7	16-3/4	---	190	---	---	---	---	---	---	135
8	18-1/2	11-1/2	275	---	---	---	---	5	82	174
1/ 9	20-1/4	---	360	---	---	---	---	---	---	206
10	21-3/4	13-1/4	330	---	---	---	---	5-3/4	125	238
12	25-1/4	14-3/4	500	---	---	---	---	7-1/8	200	330
14	---	---	---	20	662	12-3/4	464	8-3/4	325	422
16	---	---	---	20-3/4	850	13	595	9-1/8	415	537
18	---	---	---	21-3/4	1030	13-1/4	721	10-3/8	555	648
20	---	---	---	26	1232	13-3/4	862	11-1/2	725	893
22	---	---	---	26-1/2	1430	14	1001	---	---	---
24	---	---	---	28	1633	14-1/4	1143	12-1/2	1100	1500
1/ 25	---	---	---	28-1/2	1675	14-1/4	1172	---	---	---
26	---	---	---	30	1730	14-1/2	1211	---	---	---
1/ 27	---	---	---	30-1/2	1800	15	1260	---	---	---
28	---	---	---	31	1850	16	1295	---	---	---
1/ 29	---	---	---	32-1/4	1900	16	1330	---	---	---
30	---	---	---	34	1923	16-1/4	1346	---	---	---
32	---	---	---	34-3/4	2150	16-1/2	1505	---	---	---
34	---	---	---	36-1/4	2350	17-1/2	1645	---	---	---
36	---	---	---	38	2598	18	1818	---	---	---
38	---	---	---	39-3/4	2900	19	2030	---	---	---
40	---	---	---	40-3/4	3130	20	2191	---	---	---
2/ 42	---	---	---	41-1/2	3430	22	2403	---	---	---
46	---	---	---	44-1/2	3855	24-1/4	2800	---	---	---

1/ For replacement purpose only. Not to be used for new construction.
2/ Sizes 42 inches and larger shall be as approved.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

3.4.10.3 The type E valves shall be of an elliptical or rectangular top-entry with a bonnet cap which will allow replacement of the disc and seat without the need for removal of the valve body from the piping system.

3.4.11 Type G and H valves (split disc check).

3.4.11.1 Plate springs for sea water applications shall be in accordance with ASTM B 443.

3.4.11.2 Type G and H valves shall be constructed to be assembled with ANSI B16.24 flanges.

3.4.11.3 For type H valves, lugs shall be either threaded or clear drilled as specified (see 6.2.1).

3.4.11.4 Thread sealant. Thread sealant for threaded parts, except where silver brazing alloy is applied, shall be sealing compound in accordance with MIL-S-45180, type II. A very thin coat of sealing compound shall be applied to threads and bearing surfaces. After installation, remove any excess sealing compound. The sealant compound may also be used to seal gasket at body and cap joint. No other gasket sealing compound is permitted.

3.4.11.5 Lubrication. Lubrication of valve parts shall be graphite in isopropanol (Military symbol CGI) in accordance with MIL-L-24131.

3.4.12 Welding. Unless otherwise specified (see 6.2.1), weld repair and inspection of nickel-aluminum-bronze castings shall be in accordance with MIL-STD-278.

3.5 Shock resistance. Valves shall withstand high impact shock in accordance with grade A, hull-bulkhead mounted principal unit, complete assembly, class I, type A of MIL-S-901 and MIL-STD-798. Valves shall be hydrostatically pressurized internally to the maximum gauge pressure as specified in table II (see 3.6). Testing medium shall be clean tap water with no additive other than cutting oil at temperatures not less than 40°F nor exceeding 150°F. Momentary malfunction at the time of impact blow is permitted and acceptable. Permanent deformation, misalignment or functional impairments shall be cause for rejection. Permissible seat leakage shall be as specified in 3.7.

3.6 Hydrostatic pressure. Valves shall withstand hydrostatic pressure of 150 percent of the valve rated pressure (at 150°F) as specified in table II.

3.7 Seat leakage. Seat leakage rates for the following valve sizes shall not exceed 25 cubic centimeters (cm^3) per hour per inch of size. When specified (see 6.2.1) where nominal leakage cannot be tolerated, seat leakage shall not exceed 10 cm^3 per hour per inch of nps.

3.8 Vibration resistance. The valves shall withstand vibration in accordance with type I of MIL-STD-167-1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

3.9 Body marking and identification plates.

3.9.1 Body marking. The following markings shall be integrally cast on the valve body:

(a) Manufacturer's name or trademark.
(b) Valve size in inches.
(c) Valve pressure rating in lb/in^2.
(d) MIL-V-17547.
(e) Direction of flow arrow.

3.9.2 Identification plate. Valves shall be provided with an identification plate in accordance with MIL-P-15024 and MIL-P-15024/5, type A, B, C or D. Identification plates made of corrosion-resisting steel or brass shall be permanently fastened to a part of the valve not subjected to working pressure. The identification plate shall contain the following:

(a) Manufacturer's name or trademark.
(b) Type, class and size of valve in inches and rating in lb/in^2.
(c) Body and material composition.
(d) Valve trim material (disc, seat, and so forth).
(e) Manufacturer's identification number (Commercial and Government Entity (CAGE)).
(f) MIL-V-17547.
(g) Component identification number (CID) (if known).
(h) Contract number.
(i) Functional group code number (FGC) (if known).

3.9.2.1 Order of precedence. Identification plates may be reduced in size for the smaller valves by limiting the contents to items (b), (c) and (g). Identification plates, when attached, shall not cover the integrally cast marking.

3.10 Flanged valves. Dimensions, weights, and flow coefficient (C_v) shall be as specified in 3.10.1 through 3.10.2.

3.10.1 Dimension and weights for valves 2-1/2 inches and larger shall be as specified in table III.

3.10.2 The C_v for valves shall be as specified in table IV. The flow rates are based on values for valves in the fully open position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

TABLE IV. Minimum C_v.

Valve size (nps)	Type A Swing check, class 2, 3 and 4	Types B, C, D Lift check, class 2, 3 and 4			Types E, F Tilting disk check, class 1	Types G and H
		Horizontal	Vertical	Angle		
1/4	4	1	3	2	---	---
3/8	7	2	6	3	---	---
1/2	12	3	11	6	---	---
3/4	20	6	19	10	---	---
1	35	10	33	17	---	---
1-1/4	61	17	58	30	---	---
1-1/2	82	23	78	41	---	---
2	139	40	133	69	---	---
2-1/2	214	61	204	107	---	82.1
3	316	91	302	158	---	139.3
1/ 3-1/2	423	122	403	211	---	---
4	543	156	518	271	---	271.3
1/ 4-1/2	684	197	652	342	---	---
5	877	253	836	438	---	482.5
6	1287	371	1227	643	---	725
1/ 7	1719	456	1639	859	---	---
8	2255	651	2150	1127	---	1509
1/ 9	2859	825	2726	1429	---	---
10	3566	1029	3400	1783	---	2640
12	5200	1501	4958	2600	---	4075
14	---	---	---	---	7200	5303
16	---	---	---	---	10000	7340
18	---	---	---	---	12300	9940
20	---	---	---	---	16000	12960
22	---	---	---	---	19500	---
24	---	---	---	---	23500	20500
1/25	---	---	---	---	24750	---
26	---	---	---	---	27000	---
1/27	---	---	---	---	29250	---
28	---	---	---	---	32000	---
1/29	---	---	---	---	34000	---
30	---	---	---	---	38000	---
32	---	---	---	---	43500	---
34	---	---	---	---	49750	---
36	---	---	---	---	56250	---
38	---	---	---	---	62400	---
40	---	---	---	---	68000	---
42	---	---	---	---	69000	---
46	---	---	---	---	80874	---

1/ For replacement purpose only. Not to be used for new construction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

3.11 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2 and appendix A).

3.12 Special tools. Valves shall be constructed so that special tools are not required for installation and maintenance. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)).

3.13 Workmanship. All parts shall be free of adhering sand, hard spots, excessive shrinkage, cracks or scale. Unacceptable visual surface discontinuities such as cracks, tears, laps, sand inclusions and gas holes shall be removed and their removal shall be verified by visual examination. Casting shall not be repaired, plugged, impregnated, brazed, burned in or peened, except as specified in 3.4.12.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order, an inspection system program plan shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5 and 4.6. One valve of each type and class shall be subjected to first article inspection. Acceptance of the inspected sizes shall include the sizes not inspected as shown in table V. First article tests shall be monitored by a Government representative. The contractor

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

shall not proceed with production under the contract or order without submitting all approved test data results certified by the Government representative and without receiving final written approval from the contracting activity (see 6.3.1).

TABLE V. Valve test sizes.

Size tested (inches)	Approves these sizes (inches)
3/4	3/4 and smaller
2	1 to 2, inclusive
6	2-1/2 to 6, inclusive
8	7 and 8
10	9 and 10
over 10	All sizes shall be tested

4.3.1 First article inspection procedure and report. When specified in the contract or order, a first article inspection procedure and report shall be prepared (see 6.2.2 and appendix C).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in 4.5 and 4.6.4. Quality conformance inspection shall be performed on each valve acquired under this specification.

4.4.1 Sampling.

4.4.1.1 Lot. A lot shall consist of all valves of the same type, class, and size offered for delivery at one time for the purpose of examination and tests.

4.4.1.2 Sampling for visual and dimensional examination. Sample valves shall be selected at random from each lot of completely finished valves in accordance with MIL-STD-105 at inspection level II. For lots of 180 and under, samples shall be as specified in table VI.

TABLE VI. Sampling for visual and dimensional examination.

Lot size	Sample size	Acceptance number (defective)
2 to 8	2	0
9 to 15	8	0
16 to 25	10	0
26 to 65	16	0
66 to 180	26	0

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

4.5 Examinations.

4.5.1 Examination of materials. Materials used in the manufacture of valves shall be examined to determine conformance to 3.2 and table I, as applicable.

4.5.2 Visual and dimensional examination. Each sample valve shall be examined by the inspector to determine compliance with all requirements of the specification not involving tests. Each sample valve that fails to meet these requirements shall be rejected. If the number of nonconforming valves in any sample exceeds the acceptance number for that sample size, the entire lot represented by the sample shall be rejected. Rejected lots may be resubmitted for examination after the contractor has inspected an additional sample (same sample size) of the remaining uninspected valves. Acceptance number for this second sampling shall be the same as the first. Failure of this second examination shall require the contractor to examine all remaining uninspected valves at the same acceptance number. Failure of this examination will form the basis for cancelling the purchase.

4.6 Tests.

4.6.1 Shock. The valve shall be subjected to the grade A shock test in accordance with MIL-S-901 to determine conformance to 3.5.

4.6.2 Vibration. The valve shall be subjected to the type I vibration test in accordance with MIL-STD-167-1 to determine conformance to 3.8.

4.6.3 Operational tests. Operational tests shall be performed on each valve size and type. Operational tests shall be conducted in accordance with the test conditions in 4.6.4.1 alternately applying the pressure specified in 3.6 with the pressure on the seat with the valve in the check position for 2000 cycles. One cycle shall consist of the above specified pressure applied in the check position and the open position. At the completion of the operational test the seat tightness test of 4.6.4.3 shall be performed.

4.6.4 Hydrostatic pressure and seat leakage.

4.6.4.1 Test conditions. Each valve shall be hydrostatically tested for structural integrity and seat leakage using commercially available tap water at temperatures not less than 40°F and not more than 150°F, and shall contain no additives other than cutting oil. Test fixtures shall be adjusted to permit valve body expansion and elongation due to internal pressure to the fullest extent possible. The thrust on the ends shall be the minimum required to facilitate testing and prevent leakage to the atmosphere.

4.6.4.2 Hydrostatic pressure test method. Each valve shall be hydrostatically pressurized internally in the open position to 150 percent of the valve rated pressure specified in table II. Pressure shall be maintained until a thorough visual inspection has been conducted but not less than 1 minute. Any weeping or leakage during pressurizing or permanent deformation after pressurizing shall be cause for rejection. The appearance of water or wetting at the fluid boundary formed by the gasket is cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

4.6.4.3 Seat tightness test. Each valve shall be subjected to the test pressure, as specified herein, with the disc in the closed position. If no leakage occurs after 1 minute of pressurization, the test may be discontinued. If there is any visible leakage, the test shall be continued for a sufficient length of time to accurately determine the rate of leakage. Leakage shall not exceed that specified in 3.7.

Valve primary pressure rating	Test in check position
150 lb/in^2 and below	50 lb/in^2
Over 150 lb/in^2	100 lb/in^2

4.6.4.4 Acceptance criteria. Any valve which does not meet the requirements for hydrostatic pressure and leakage shall be rejected.

4.6.5 Silver brazing. Silver brazing shall be inspected in accordance with NAVSEA 0900-LP-001-7000 to determine conformance to 3.4.2.

4.6.6 Repair by welding. Repairs shall be made using procedures and welder qualifications in accordance with 3.4.12. Weld repairs shall be inspected in accordance with MIL-STD-278.

4.7 Test reports. When specified in the contract or order, test reports shall be prepared (see 6.2.2).

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 of this specification and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation-packaging, packing and marking. Valves shall be individually preserved-packaged level A or C, packed level A, B or C as specified (see 6.2.1) and marked in accordance with MIL-V-3.

5.2 Cushioning, dunnage and wrapping materials.

5.2.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler or dunnage is prohibited for materials destined for shipboard installation or stowage.

5.2.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging and packing applications such as cushioning, filler and dunnage, all containers (unit, intermediate and shipping) shall be marked or labeled with the following information:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

"CAUTION

Contents cushioned etc., with loose-fill material. Not to be taken aboard ship. Remove and discard loose-fill material before shipboard stowage. If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material."

5.2.3 Cushioning, filler, dunnage, and wrapping materials selected shall have properties (characteristics) resistant to fire.

6. NOTES

6.1 Intended use. Types A, B, C and D valves are intended for use primarily for water, oil or gas service. Types E and F valves are intended for use on seawater services as main injection and overboard discharge valves.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Functional group code number, if known.
(c) Type, class and size (see 1.2).
(d) Quantity.
(e) Material composition if other than specified (see 3.2).
(f) If valve sizes 2 inches and below should be furnished other than specified (see 3.4.5).
(g) Flange drilling if other than specified and if odd number of bolt holes is required (see 3.4.6).
(h) For type H valves, if lugs are to be threaded or clear drilled (see 3.4.11.3).
(i) Welding if other than specified (see 3.4.12).
(j) When seat leakage other than nominal is required (see 3.7).
(k) Level of preservation-packaging and packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.11 and appendix A	Engineering drawings	DI-DRPR-80651	Level 3
4.1.2	Inspection system program plan	DI-R-4803	7027 2398
4.3.1	First article inspection procedure	DI-T-4901	----
3.2, 4.3.1 and appendix C	Test reports	DI-MISC-80653	----
4.7	Reports, test	DI-T-2072	10.1.b

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for an identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract. Technical content shall include the requirements of the appendix B, titled "Manual Technical Content Requirements".

6.3 First article inspection. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1 For those products requiring first article inspection, bidders are required to contact their Government inspector to make arrangements for monitoring the tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Valve, lift angle
Valve, lift horizontal
Valve, lift vertical
Valve, swing
Valve, tilting disc (in line)
Valve, tilting disc (top entry)

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4820-N518)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

20.1 Other publications. The following documents form a part of this appendix to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1 - Surface Texture (Surface Roughness, Waviness and Lay). (DoD adopted)

Y14.5 - Dimensioning and Tolerancing. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

30. DRAWINGS

30.1 Drawings. When required by the contract or order, drawings shall contain the following information:

(a) Drawings showing cross-sectional assembly which clearly depicts the design and construction of the valve.

(b) Bill of materials listing specification, grade, condition, and any other data required to fully identify the properties of the materials proposed.

(c) Detail drawings of all parts and subassemblies necessary for evaluation of the equipment and parts necessary for maintenance and overhaul of the valve. Details of these parts should be so complete as to permit emergency manufacture by a Naval shipyard without assistance from the original manufacturer. Subassembly parts which cannot be acquired or serviced individually should be shown as a single part and so indicated. Multidetail drawings are preferred, but monodetail drawings may be used.

(d) Recommended assembly torques or equivalent procedures, for making up all joints and threaded assemblies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

(e) Tabulation of required gasket characteristics including all dimensions (with tolerances) and load versus compression characteristics (with tolerances).
(f) Dimensions-overall in accordance with ANSI Y14.5, accessibility space including disassembly clearances and all dimensions pertinent to installation.
(g) Surface finishes in accordance with ANSI B46.1 for all hard face surfaces and bearing areas.
(h) Name of laboratory conducting tests and date of previous successful shock and vibration tests.
(i) Calculated weight.
(j) Valve with weights of 100 pounds or greater should indicate the vertical, longitudinal, and transverse location of the center of gravity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

APPENDIX B

MANUALS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included in technical manuals when required by the contract or order. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. MANUALS

30.1 Manuals. When required by the contract or order, manuals should contain the following information:

(a) Drawings of the valve including certification data sheet. Drawings should be supplemented by additional illustrations where necessary to adequately illustrate operation and maintenance. These additional illustrations may consist of exploded views, partial or full sections, and may eliminate extraneous lines and details to clarify the interaction of parts.

(b) Detailed disassembly and reassembly procedures. In addition to providing procedures for the complete disassembly and reassembly of the equipment, maintenance and troubleshooting sections should contain, or refer to, only the limited disassembly and reassembly required to accomplish each particular operation. This is intended to reduce the possibility of unnecessary disassembly and unnecessary disturbance of adjustments when performing specific or limited maintenance or troubleshooting operations.

(c) Tables listing wrench sizes and torque values or other equivalent procedures for assembling joints and threaded parts.

(d) Instructions to permit overhaul by shipyard or other repair facility. These should include procedures for inspecting critical dimensions subject to wear or change and the acceptable dimensional limits, surface finish condition, and other necessary characteristics. Also, the appropriate procedure (that is, part replacement, correction at repair facility, or repair at manufacturer's facility) which should be followed to correct each case of damage or wear.

(e) Adjustment procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

APPENDIX C

TEST REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included in certificates of compliance and first article inspection reports when required by the contract or order. This appendix is mandatory only when data item description DI-MISC-80653 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. REPORTS

30.1 Certificate of compliance. When required by the contract or order, certificates of compliance should contain the following information:

(a) Valve specification.
(b) Valve type.
(c) Valve size.
(d) Valve style.
(e) Valve class.

30.2 First article inspection report. When specified in the contract or order, first article inspection reports should contain the following information:

(a) Valve, check.
(b) MIL-V-17547.
(c) Type and class.
(d) Manufacturer's name.
(e) Component identification number (if known).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-17547A(SH)	VALVES, CHECK, BRONZE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:16Z
Check the source to verify that this is the current version before use.